- It is the ratio of one species population over total number of organisms across all species in the given biome. ‘Zero’ would be infinite diversity, and ‘one’ represents only one species present.
- Species diversity is a measure of the diversity within an ecological community that incorporates both species richness (the number of species in a community) and the evenness of species.
- In general, species diversity decreases as we move away from the equator towards the poles.

55.

- Genetic diversity is the total number of genetic characteristics in the genetic makeup of a species.
- A single species might show high diversity at the genetic level (For example, Homo sapiens: Chinese, Indian American, African, etc.).
- India has more than 50,000 genetically different strains of rice and 1,000 varieties of mango.
- Genetic diversity allows species to adapt to changing environments. This diversity aims to ensure that some species survive drastic changes and thus carry on desirable genes.

56.

- Keystone species is a species whose addition to or loss from an ecosystem leads to major changes in the occurrence of at least one other species.
- Certain species in an ecosystem is considered more important in determining the presence of many other species in that ecosystem.
- A keystone species is often, but not always, a predator. Just a few predators can control the distribution and population of large numbers of prey species. Herbivores can also be keystone species. Their consumption of plants helps control the physical and biological aspects of an ecosystem. Other kinds of keystone species are those, such as coral or beavers, that significantly alter the habitat around them and thus affect large numbers of other organisms.

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